

P1 MOUNTING

PRE-SPRINT MEETING

DESIGN CHANGED

CONCEPT STILL USEFUL

*This is for visualizing the mounting of the Pi-5 and touch screen display

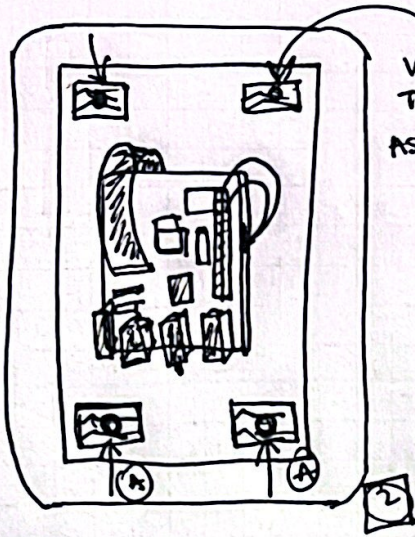
-> The mounting points that come with the display will be used in the new design for the back plate

-> The new back plate has taken a different form, so I've included this as a historical reference

WILL USE THESE HOLES AS MOUNTING POINTS

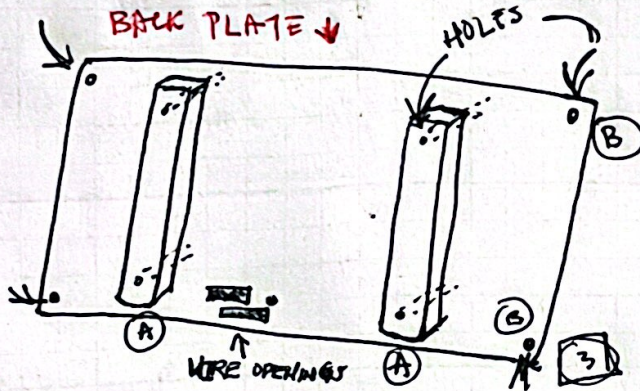
DISPLAY & P1 5

PORTRAIT =>
LANDSCAPE
(ACTUAL)



BACK PLATE

HOLES



WIRE OPENINGS

~~BACK PLATE~~

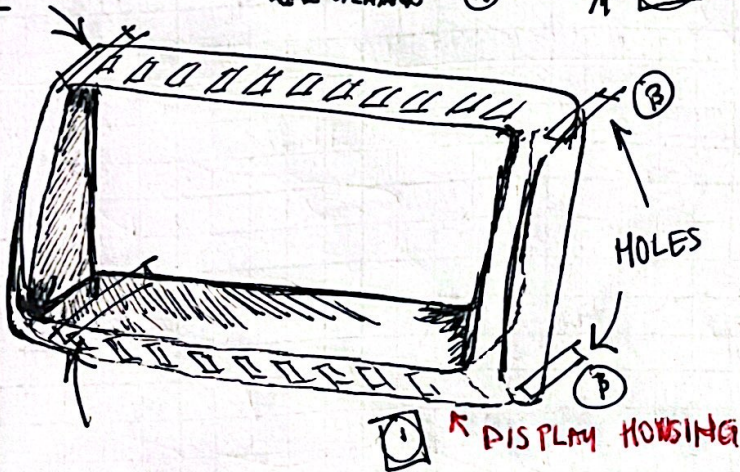
A maps to A

↓

B maps to B

* Screw for A will enter from the back & extend all the way through to display

* Screw for B will connect back plate and display housing



HOLES

DISPLAY HOUSING



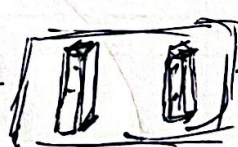
P1.a

+

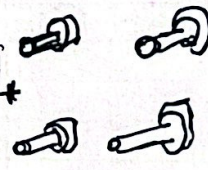


DISPLAY
+
RASPBERRY PI

+



P1.b



SCREWS

x 2

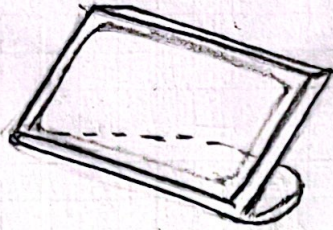
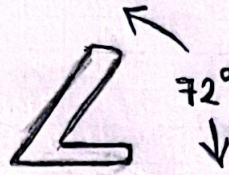
(8) screws

***POST MONDAY 03/23 SPRINT MEETING**

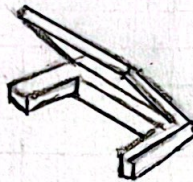
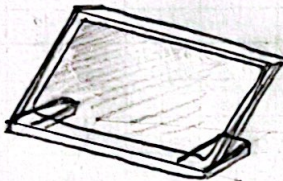
-> Design needed to change because the sensor is attached directly to Pi-5 through pins, not wires

IDEA #1
— G —

FULL
LENGTH
SEMI-CIRCLE



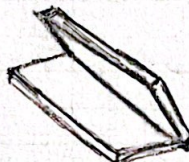
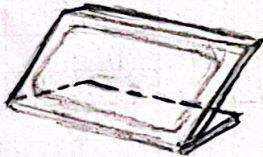
TWO
LEGS



IDEA #2

*NO SPACE FOR WEIGHTS
*FAST TO PRINT

FULL
WIDTH

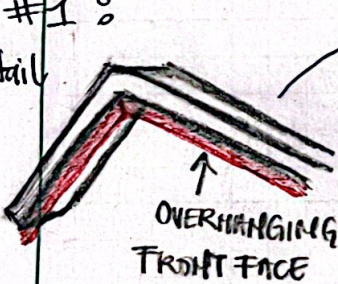


IDEA #3

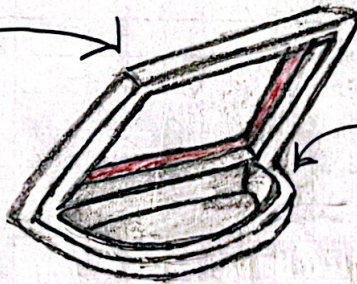
*BLOCKY

IDEA #1 %

*More detail



OVERHANGING
FRONT FACE



CAVITY FOR
WEIGHTED
BLOCKS

BACK

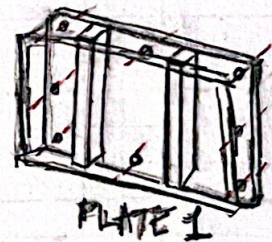


PLATE 1

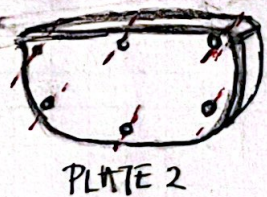
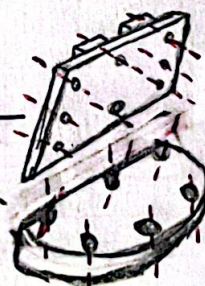
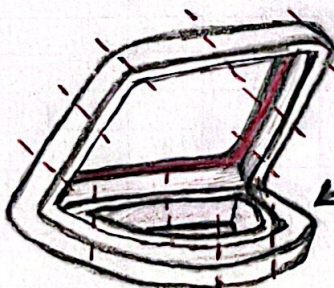
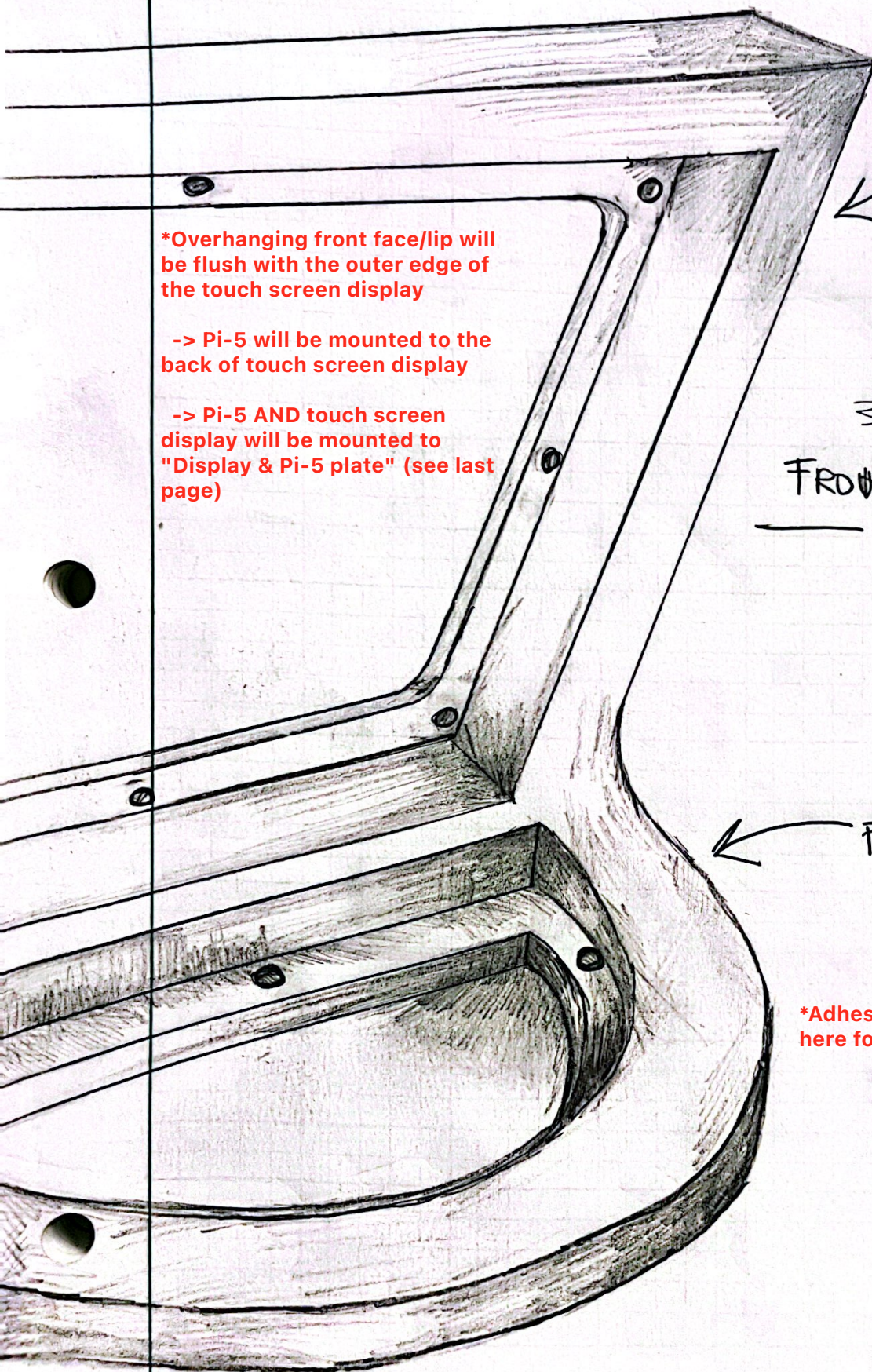


PLATE 2

*Dashes represent holes

*Preferred full-length semi-ish-circle (Idea #1)



*Overhanging front face/lip will be flush with the outer edge of the touch screen display

-> Pi-5 will be mounted to the back of touch screen display

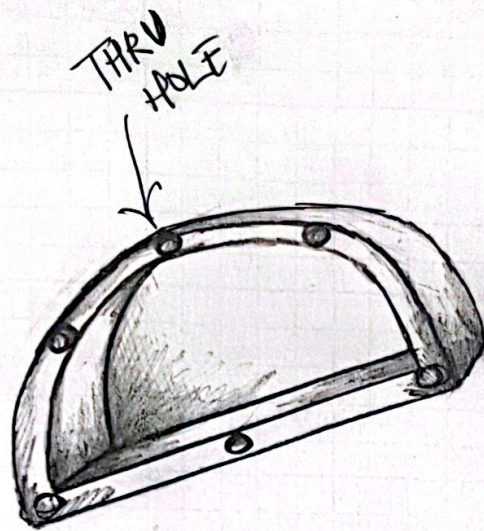
-> Pi-5 AND touch screen display will be mounted to "Display & Pi-5 plate" (see last page)

FOR DISPLAY
AND PI 5

3D-PRINTED
FRONT HOUSING

FOR WEIGHTED
BLOCKS

*Adhesive weighted blocks can go here for stability

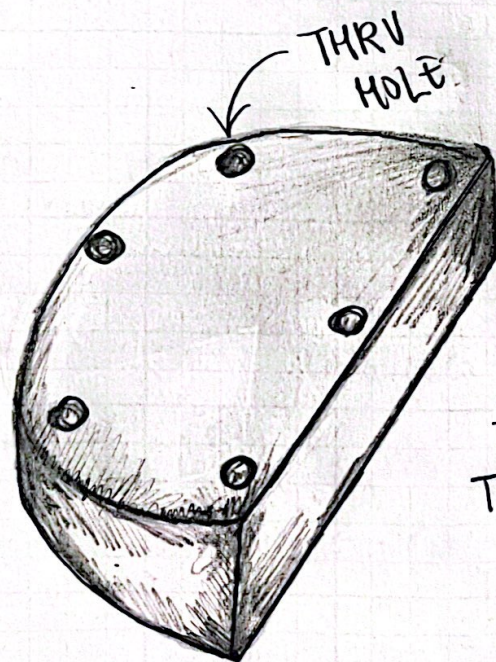


ANGLED BOTTOM
VIEW

*Internally hollow to allow for more weighted blocks, if necessary

-> Section would have to be rattle-proofed with something soft (e.g., foam, stuffing)

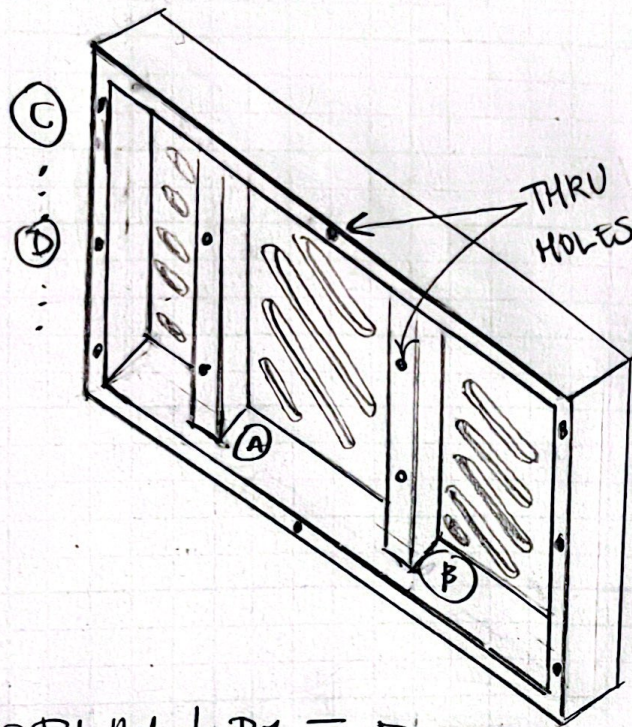
WEIGHT TRENCH
COVER



ANGLED
TOP VIEW

*Thru holes will be fastened to "3D-printed front housing" bottom section (see previous page)

*Diagonal cuts are vents



Important:

DISPLAY & P15 PLATE

- * DISPLAY'S MOUNTING BRACKETS GO ON PILLARS (A) & (B)
- * 3D-PRINTED FRONT HOUSING MOUNTS TO THIS BACKPLATE WITH HOLES (C), (D), etc
- * P15 WILL BE CONNECTED (MOUNTED) TO DISPLAY, DISPLAY & P15 WILL BE MOUNTED TO THIS, THE WHOLE THING WILL BE MOUNTED ON THE FRONT HOUSING